

Week 2
Forward Kinematics

ENN581 Robot Motion, Control and Planning

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School of Electrical Engineering and Robotics

ACKNOWLEDGEMENT OF TRADITIONAL OWNERS

QUT acknowledges the Turrbal and Yugara, as the First Nations owners of the lands where QUT now stands. We pay respect to their Elders, lores, customs and creation spirits. We recognise that these lands have always been places of teaching, research and learning.

QUT acknowledges the important role Aboriginal and Torres Strait Islander people play within the QUT community.

Last week, we learnt

- Degrees of freedom define a robot, its mobility and its workspace
- Controllable vs uncontrollable DoF
- Serial vs parallel robots in 2D and 3D

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Today's lecture is on forward kinematics

- What is forward kinematics?
- Mathematical relation between coordinate frames
- Placing coordinate frames on robots
  - Elementary Transform Sequence
  - Denavit-Hartenberg convention

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Forward Kinematics

Where is the end effector, given the coordinates of all the joints?



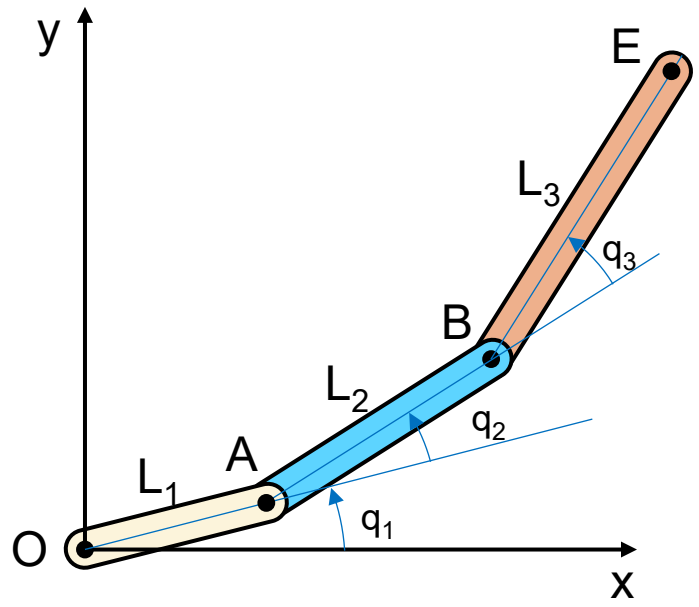
ECCE Robot

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Example: planar RRR manipulator

Where is the end effector E, given the coordinates of all the joints $\{q_1, q_2, q_3\}$?

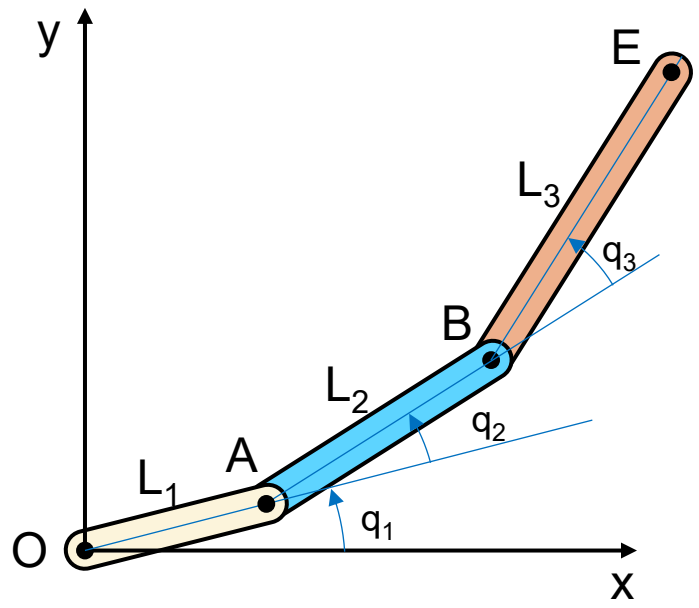
Hint: Use trigonometry / geometry to figure it out.



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Example: planar RRR manipulator

Where is the end effector E, given the position of all the joints $\{q_1, q_2, q_3\}$?



Hint: Use trigonometry to figure it out.

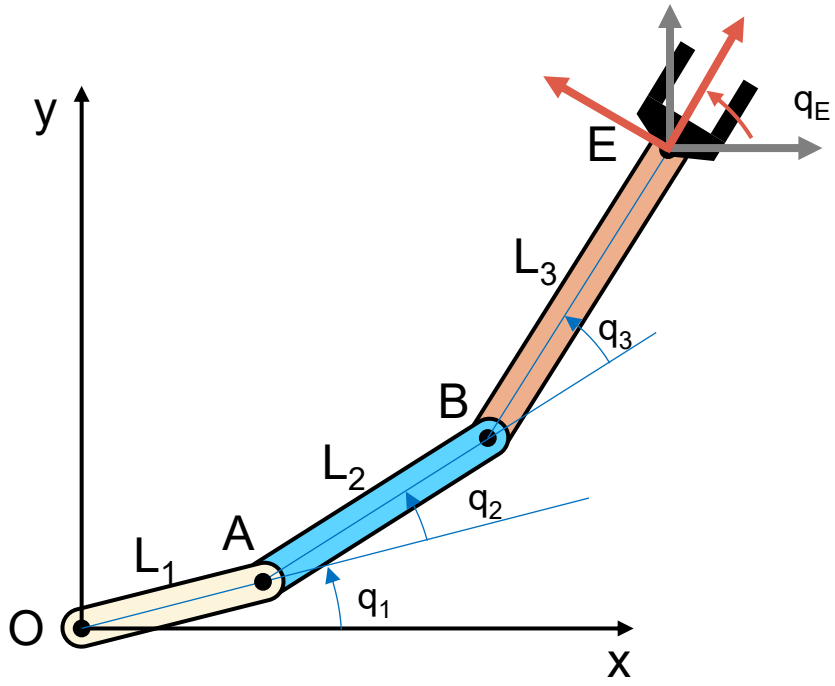
$$E_x = O_x + L_1 \cos(q_1) + L_2 \cos(q_1 + q_2) + L_3 \cos(q_1 + q_2 + q_3)$$

$$E_y = O_y + L_1 \sin(q_1) + L_2 \sin(q_1 + q_2) + L_3 \sin(q_1 + q_2 + q_3)$$

For simple kinematic chains like this, it is easy to determine the position of the end effector.

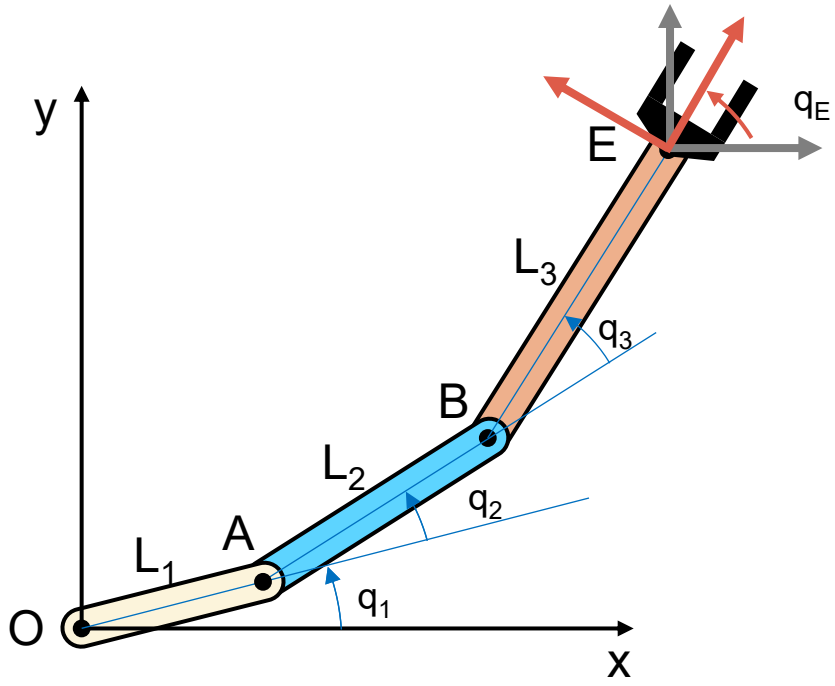
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What about the orientation?



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What about the orientation?



$$q_E = q_1 + q_2 + q_3$$

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What about manipulators with more DoF that operate in 3D space?



UR5 from Universal Robots



Panda from Franka Emika



YuMi from ABB

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Let us set up the problem Mathematically



UR5 from Universal Robots

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Homogeneous transformation matrix

Combines translation and rotation

$$ {}^{A}T_{B} = \begin{bmatrix} & {}^{A}\mathrm{R}_{B} & & {}^{A}\mathrm{t}_{B-origin} \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{A}\mathrm{R}_{B} \in SO(3) $$

$$ {}^{A}T_{B} \in SE(3) $$

$$ {}^{A}\mathrm{t}_{B-origin} \in \mathbb{R}^{3} $$

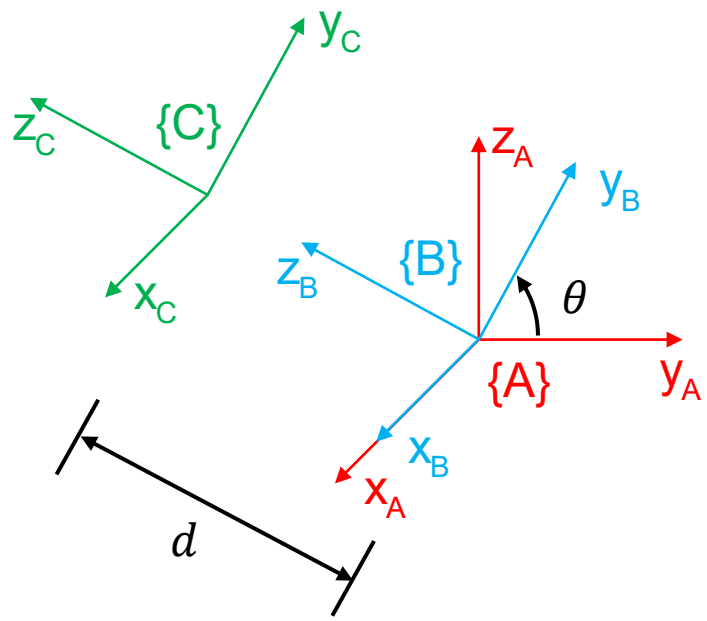
$$ {}^{A}P = {}^{A}T_{B}\,{}^{B}P $$

Homogeneous coordinates

The Homogeneous Transformation Matrix captures the relative position and orientation of two frames in a compact 4x4 matrix.

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Compound Transformations



$${}^{A}T_{B} = \mathrm{R}_{x_A}(\theta)$$

{B} is rotated by θ about x_A w.r.t. {A}

$${}^{B}T_{C} = \mathrm{t}_{z_B}(d)$$

{C} is translated by d along z_B w.r.t. {B}

$${}^{A}T_{C} = {}^{A}T_{B}\,{}^{B}T_{C}$$

Note the order in which the matrices are multiplied

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Forward kinematics of a manipulator arm can be thought of as a composition of homogeneous transformations



UR5 from Universal Robots

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Elementary Transform Sequence

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Elementary Transform Sequence

Key idea: represent the relation between two link frames in terms of a sequence of elementary transforms

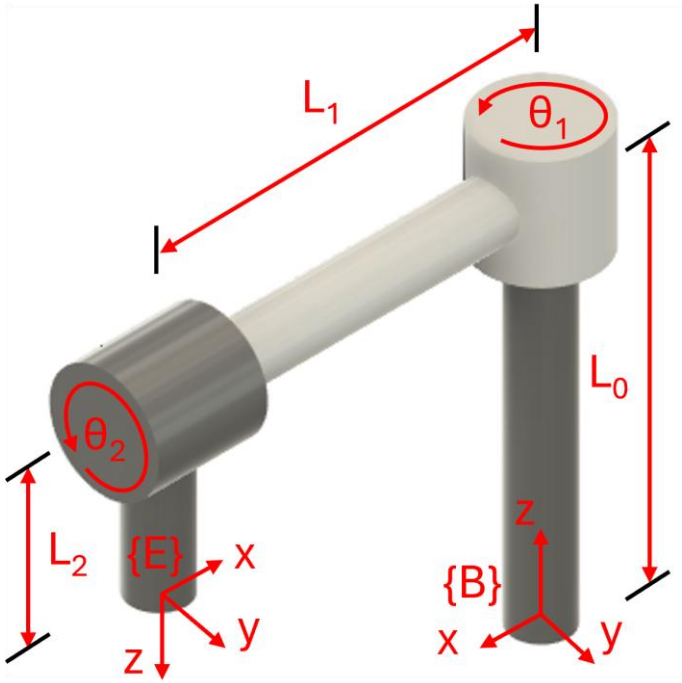
The elementary transforms are

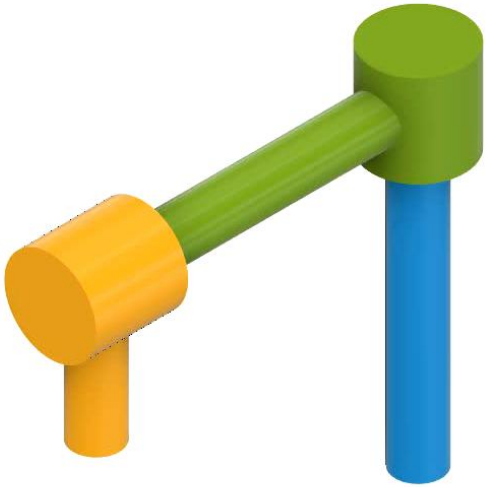
$$\mathrm{t_x}(\eta),\ \mathrm{t_y}(\eta),\ \mathrm{t_z}(\eta),\ \mathrm{R_x}(\eta),\ \mathrm{R_y}(\eta) \text{ and } \mathrm{R_z}(\eta)$$

A simple and systematic approach to assigning Denavit-Hartenberg parameters by Peter Corke
https://petercorke.com/doc/simple_systematic.pdf

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Example: Furuta Pendulum



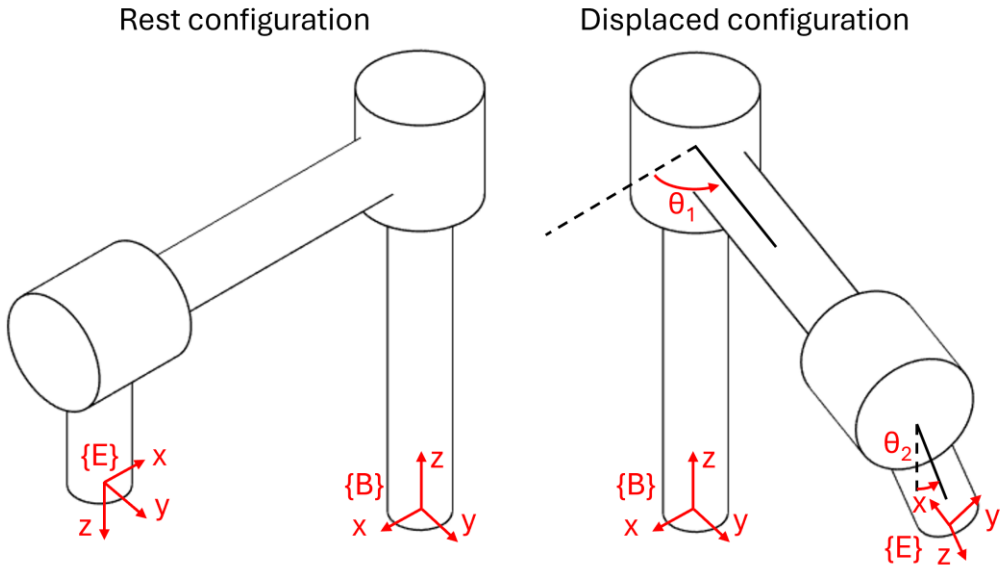


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Walk through the mechanism and assign frames

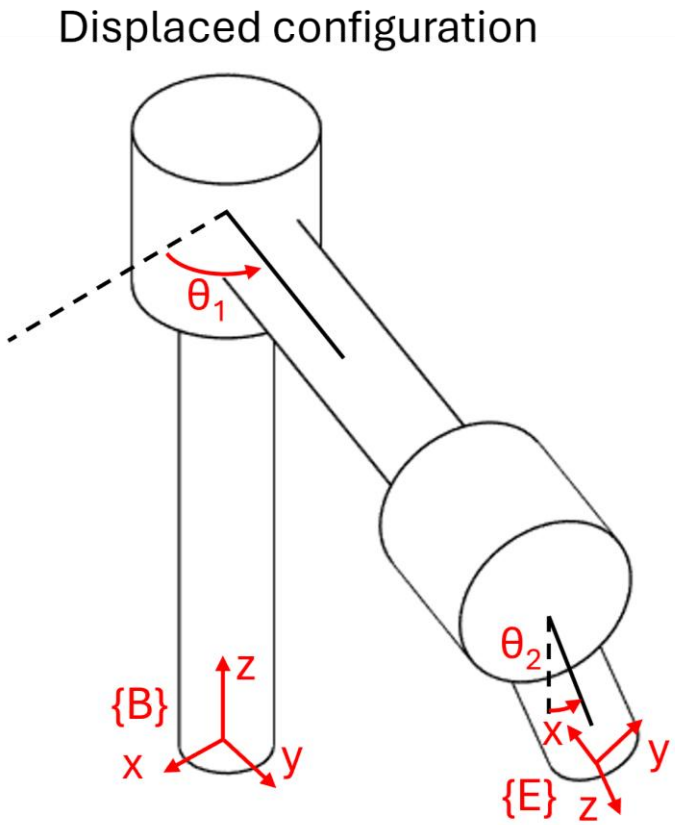
I find it useful to imagine the serial kinematic chain in a configuration in which each joint is displaced from its rest position.





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Relation between base frame and tool frame



$$^{B}T_{E} = \mathrm{t_z}(\mathrm{L_0})\mathrm{R_z}(\theta_1)\mathrm{t_x}(\mathrm{L_1})\mathrm{R_x}(\theta_2)\mathrm{t_z}(-\mathrm{L_2})\mathrm{R_y}(\pi)$$

TEQSA Provide

Denavit-Hartenberg notation for assigning frames

Motivation: There are many ways to encode kinematics using homogeneous transforms.
Sacrifice some of the flexibility for a more systematic approach.

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Proximal DH Notation

Follows the method presented by Craig in his book

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Consider an arbitrary link between two joints

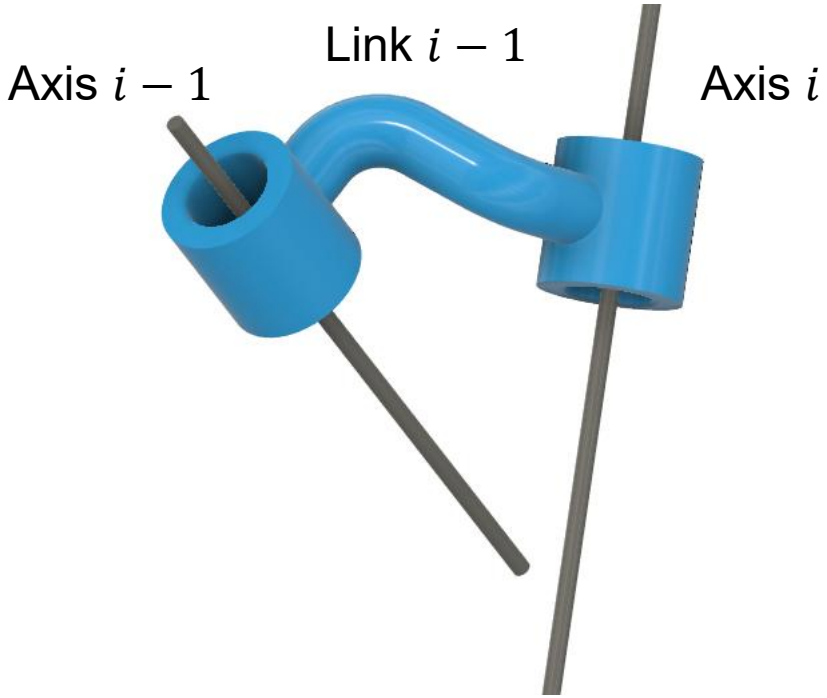
Link i-1



The shape of the link doesn’t matter

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Each joint moves about an axis

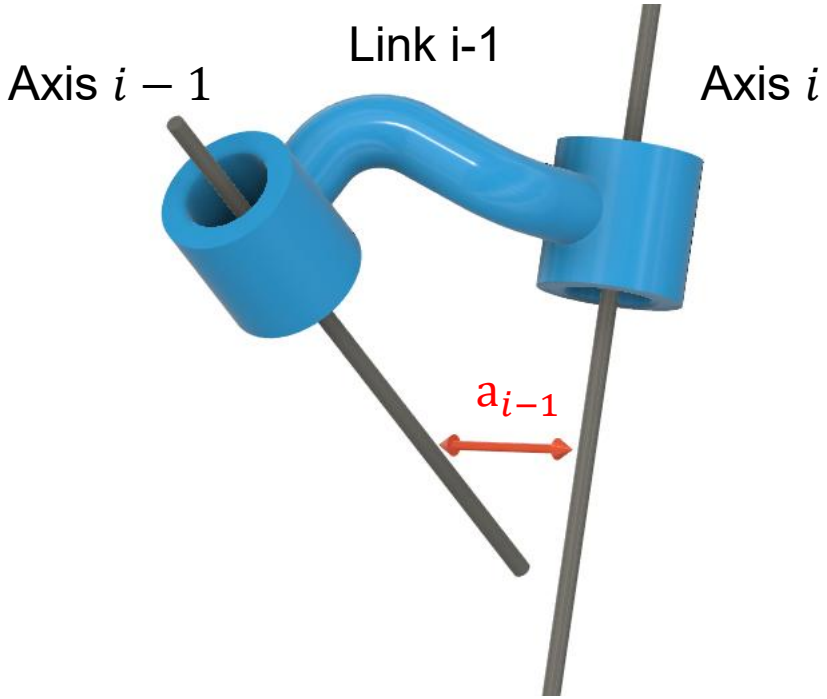


Links are numbered from 0 to n. 0 refers to the base link which is fixed.
Joint i connects link $i-1$ and link i.

The DH notation defines the relation between two adjacent link frames in terms of 4 specific parameters

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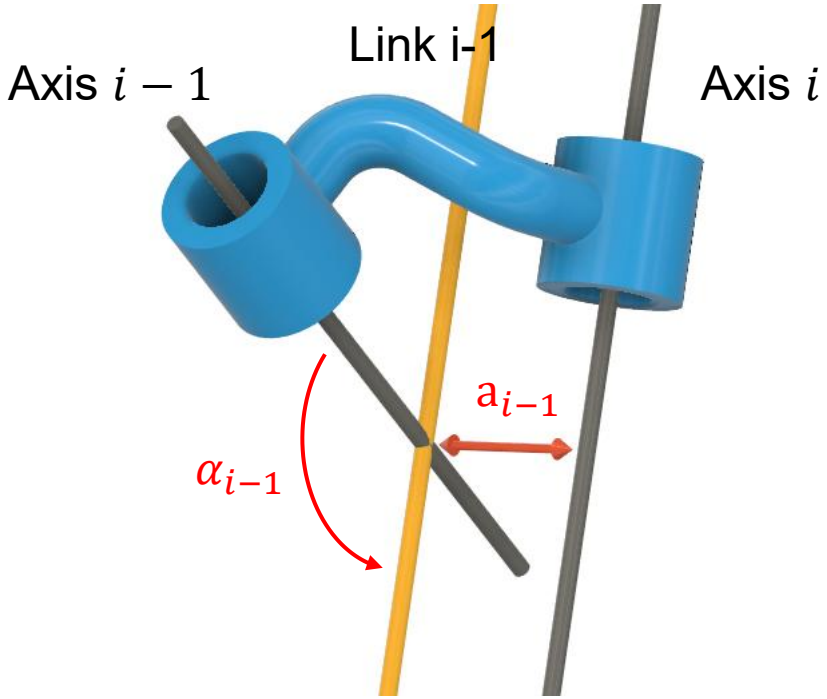
Link length: a_{i-1}



a_{i-1}: the length of the common normal between axis $i-1$ and axis i

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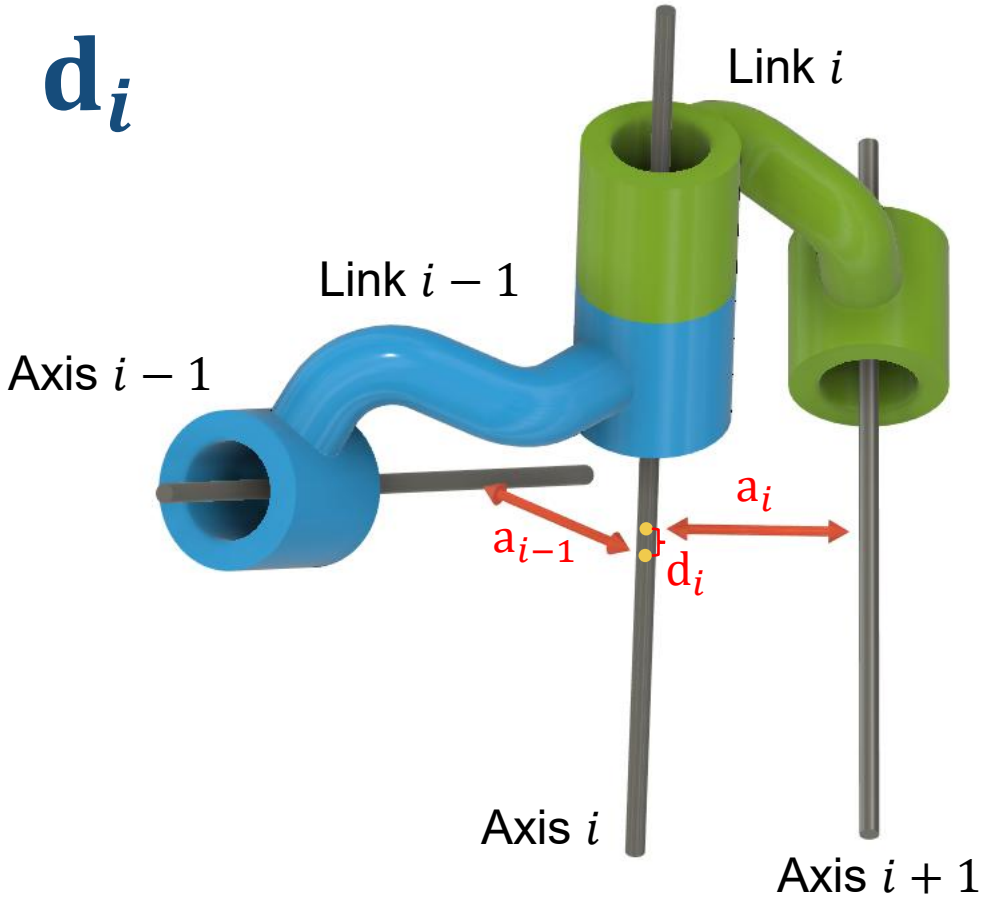
Link twist: α_{i-1}



α_{i-1}: angle measured from axis $i-1$ to the projection of axis i (in the right-hand sense) on a plane who's normal is a_{i-1}

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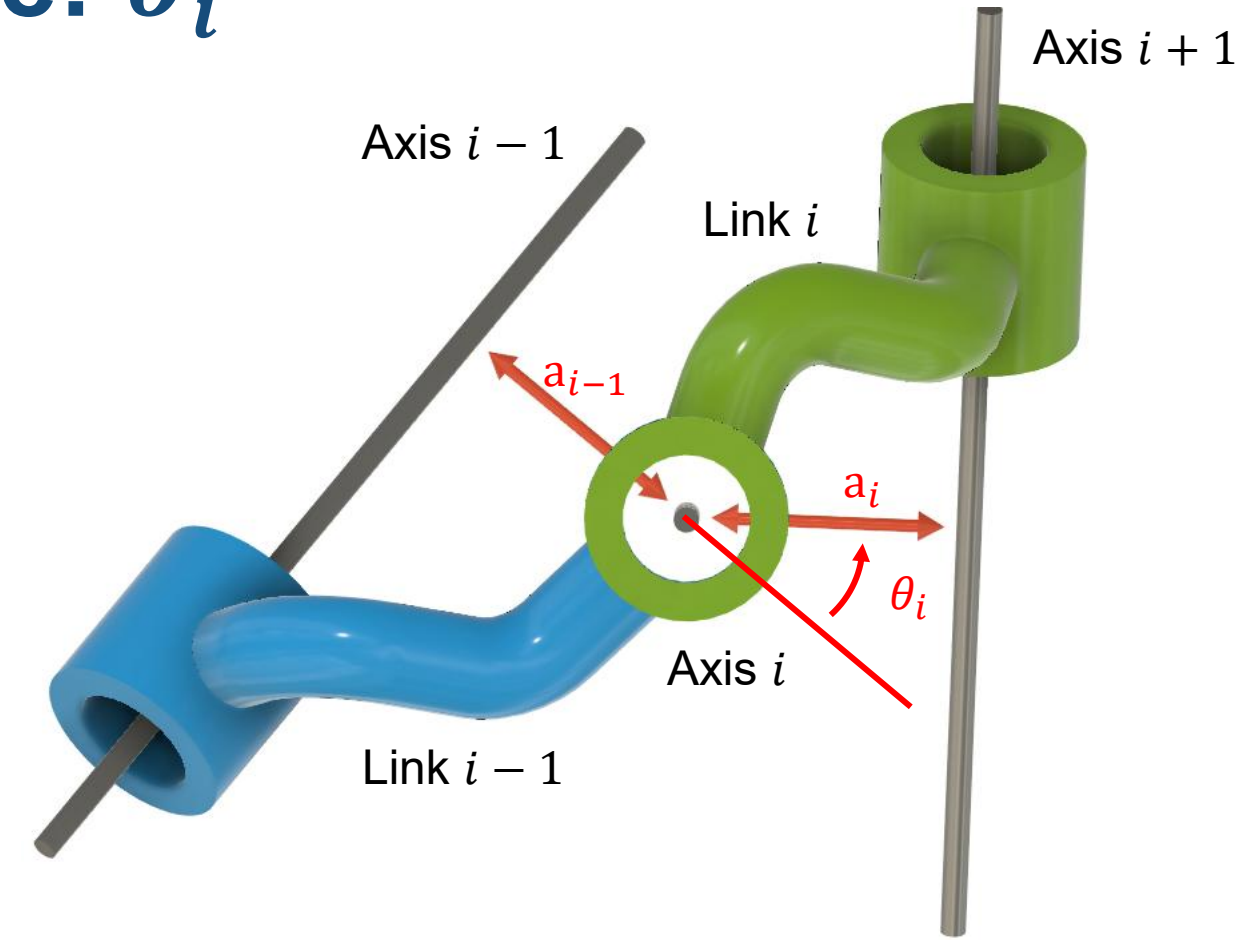
Link offset: d_i



d_i: signed distance measured along axis i from the point where a_{i-1} intersects the axis to the point where a_i intersects the axis

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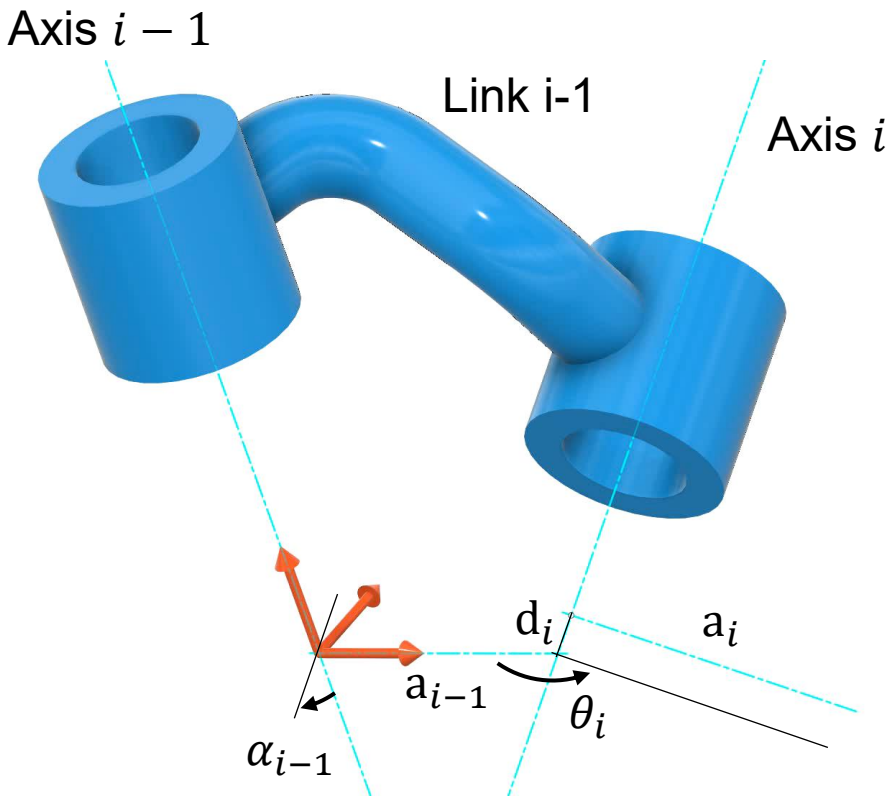
Joint angle: θ_i



θ_i: angle between an extension of a_{i-1} and a_i measured about axis i

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The two link frames are related as follows



$$^{i-1}T_i = \mathrm{R}_{x_{i-1}}(\alpha_{i-1})\mathrm{t}_{x_{i-1}}(\mathrm{a}_{i-1})\mathrm{R}_{z_i}(\theta_i)\mathrm{t}_{z_i}(\mathrm{d}_i)$$

$$^{i-1}T_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & \mathrm{a}_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}\mathrm{d}_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}\mathrm{d}_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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Assigning frames to a mechanism
(Proximal Notation or Modified DH notation)

1. Identify joint axes. For steps below, consider axes i and $i+1$.
2. Identify the common normal between them, or the point of intersection.
   - Assign the origin of link-frame i at the point of intersection of axis i and the common normal.
   - If axis i and $i+1$ intersect, then this point is the origin.
3. Assign z_i along joint axis i.
4. Assign x_i along the common normal.
   - If axis i and $i+1$ intersect, then assign x_i to be normal to the plane containing the two axes.
5. Assign y_i to complete a right-handed coordinate frame.
6. Assign $\{0\}$ to match $\{1\}$ when the first joint variable is zero.
   - For $\{N\}$, choose an origin location and x_n freely.
   - Try to cause as many linkage parameters as possible to become 0.

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The four parameters are measured as follows

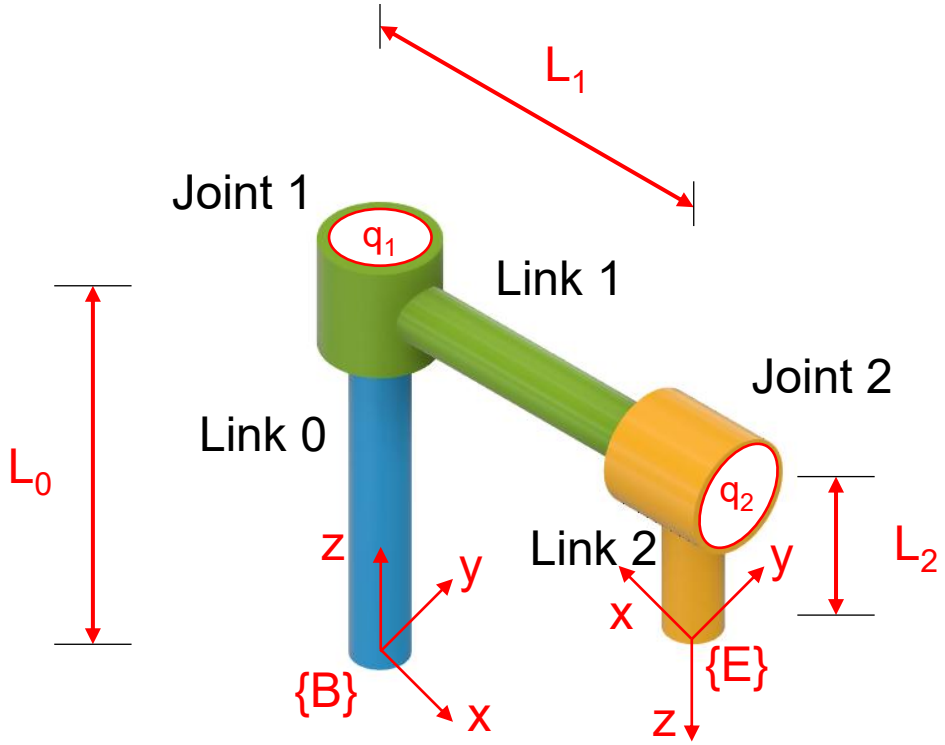
- a_i : the distance from z_i to z_{i+1} measured along x_i
- α_i : the angle from z_i to z_{i+1} measured about x_i
- d_i : the distance from x_{i-1} to x_i measured along z_i
- θ_i : the angle from x_{i-1} to x_i measured about z_i

a_i is non-negative since it is a distance. The other parameters are signed quantities.

$$
{}^{i-1}T_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & \mathrm{a}_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}\mathrm{d}_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}\mathrm{d}_i \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

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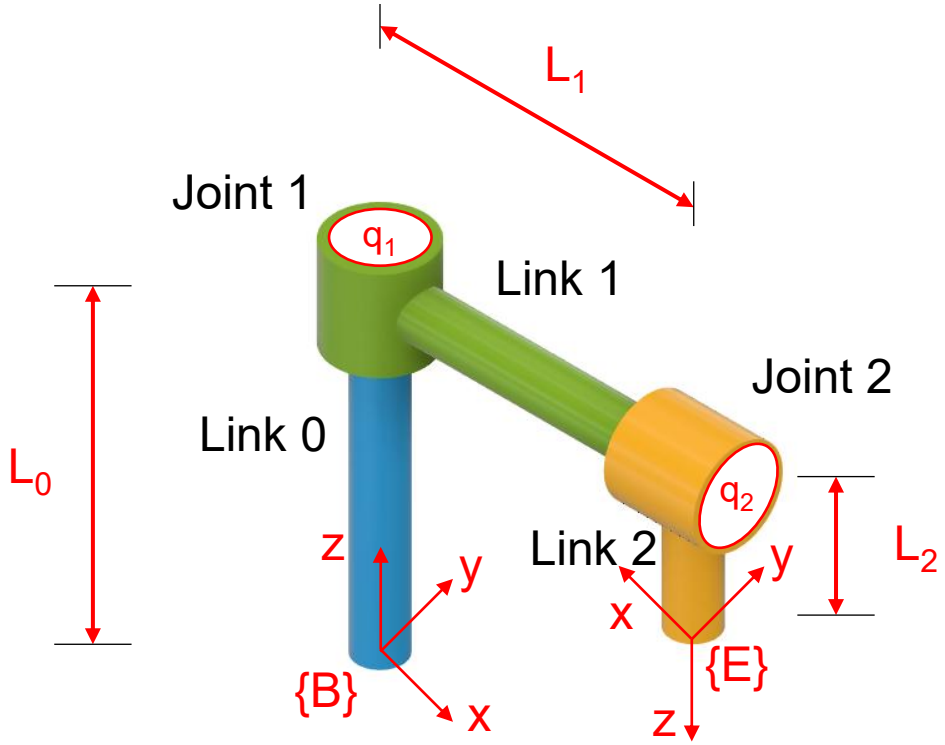
Example: Furuta Pendulum

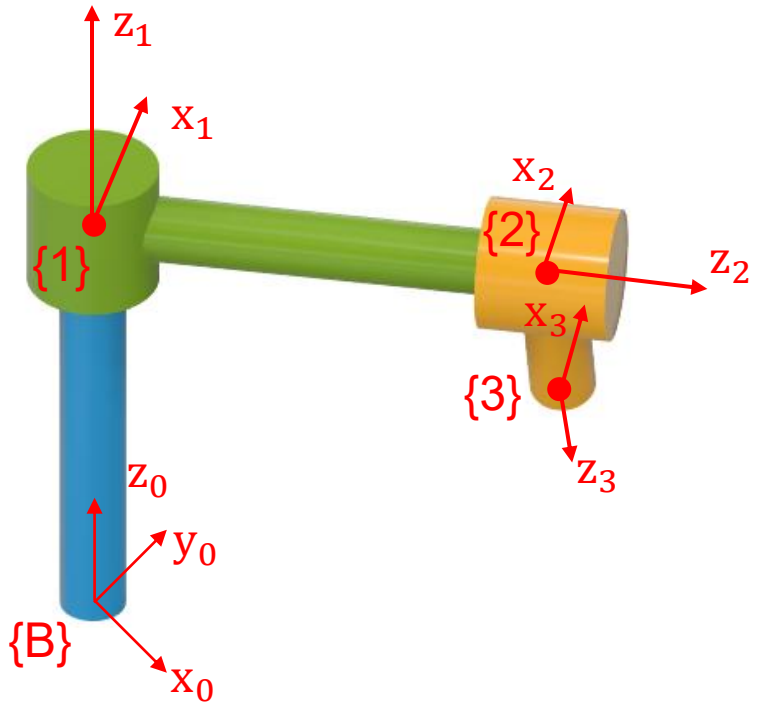


1. Identify joint axes. For steps below, consider axes i and $i+1$.
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3. Assign z_i along joint axis i.
4. Assign x_i along the common normal.
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6. Assign $\{0\}$ to match $\{1\}$ when the first joint variable is zero.
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Example: Furuta Pendulum

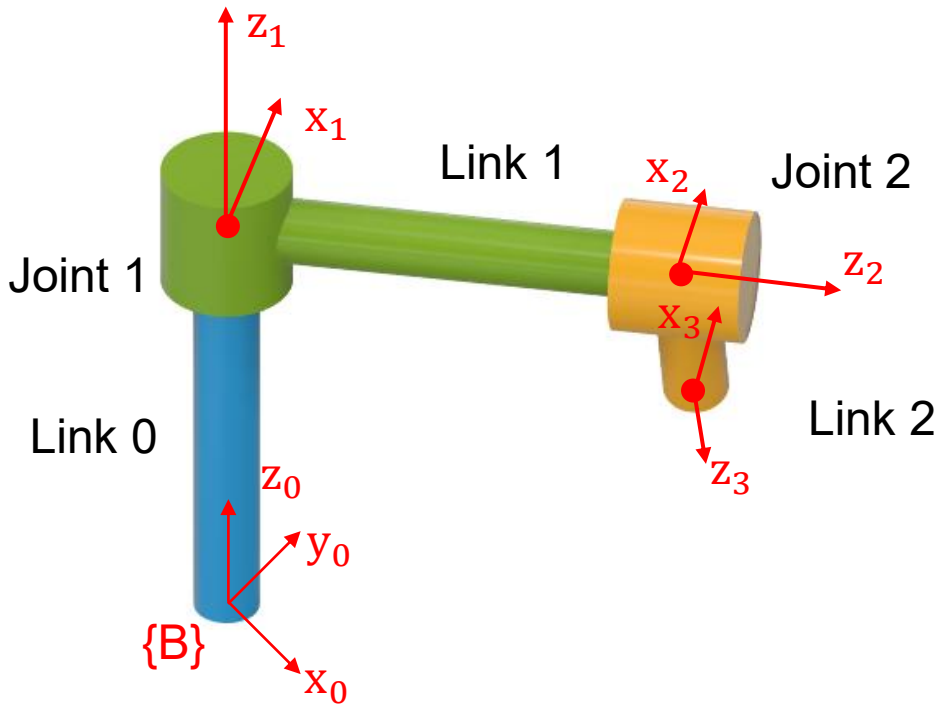




Note: Frame {3} is not the same as frame {E}
We will have to add another transformation at the end.

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Let us note down the link parameters

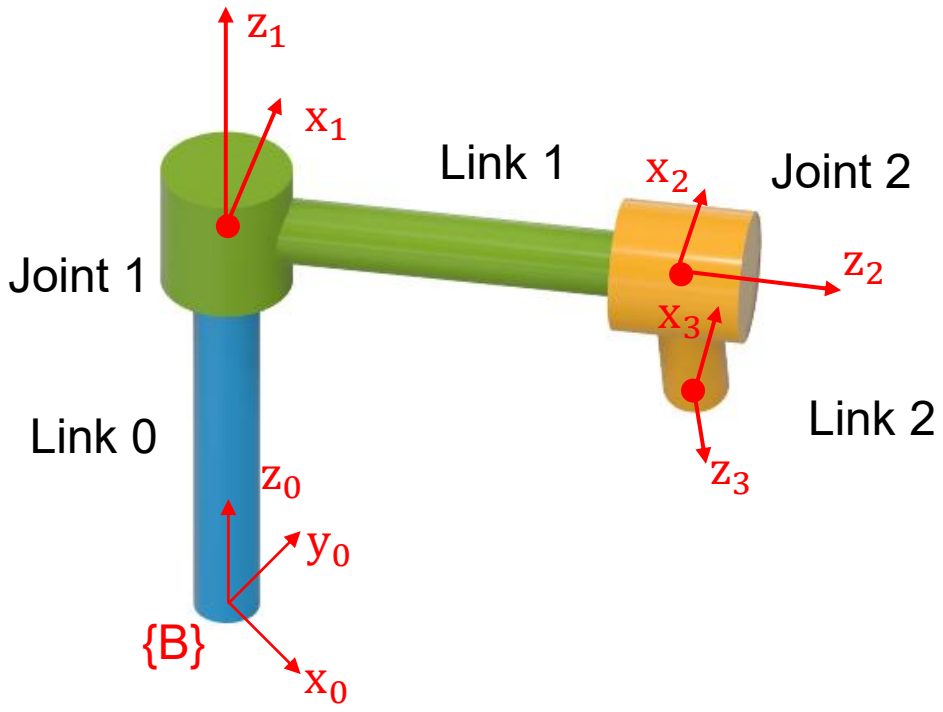


	z_{i-1} to z_i | x_{i-1}		x_{i-1} to x_i | z_i	
i	$\mathbf{a_{i-1}}$	$\boldsymbol{\alpha_{i-1}}$	$\mathbf{d_i}$	$\boldsymbol{\theta_i}$
1				
2				
3				

Note: Frame {3} is not the same as frame {E}
We will have to add another transformation at the end.

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Let us note down the link parameters



	z_{i-1} to z_i | x_{i-1}		x_{i-1} to x_i | z_i	
i	$\mathbf{a_{i-1}}$	$\boldsymbol{\alpha_{i-1}}$	$\mathbf{d_i}$	$\boldsymbol{\theta_i}$
1	0	0	L_0	q_1+90°
2	0	90°	L_1	q_2
3	0	90°	L_2	0°

Note: Frame {3} is not the same as frame {E}
We will have to add another transformation at the end.

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Relation between links in terms of parameters

The whole point of assigning frames is to get a relation between the frames of two adjacent links.

$$ {}^{i-1}T_i = \mathrm{R}_{\mathrm{x}_{i-1}}(\alpha_{i-1})\mathrm{t}_{\mathrm{x}_{i-1}}(\mathrm{a}_{i-1})\mathrm{R}_{\mathrm{z}_i}(\theta_i)\mathrm{t}_{\mathrm{z}_i}(\mathrm{d}_i) = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & \mathrm{a}_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}\mathrm{d}_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}\mathrm{d}_i \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Multiply consecutive transformation matrices to obtain ${}^{0}T_n$

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Homework:

- Use the DH parameters table for the Furuta pendulum and compute the transformation matrix relating $\{B\}$ and $\{E\}$.
- Verify that it is the same as the one we obtained using ETS.

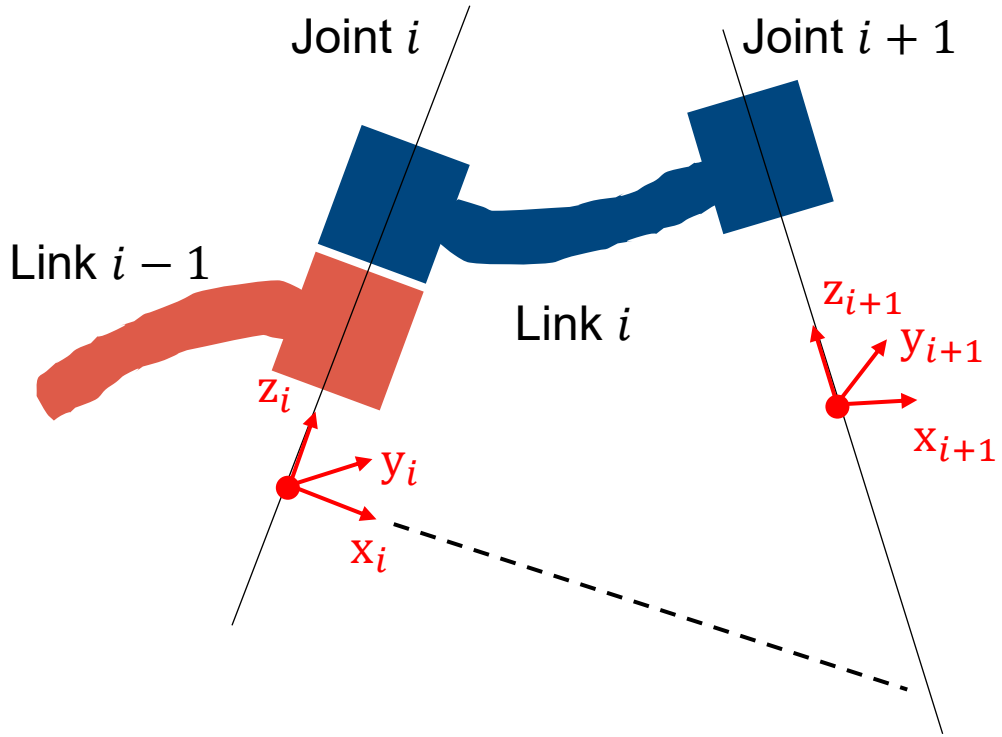
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Distal DH Notation

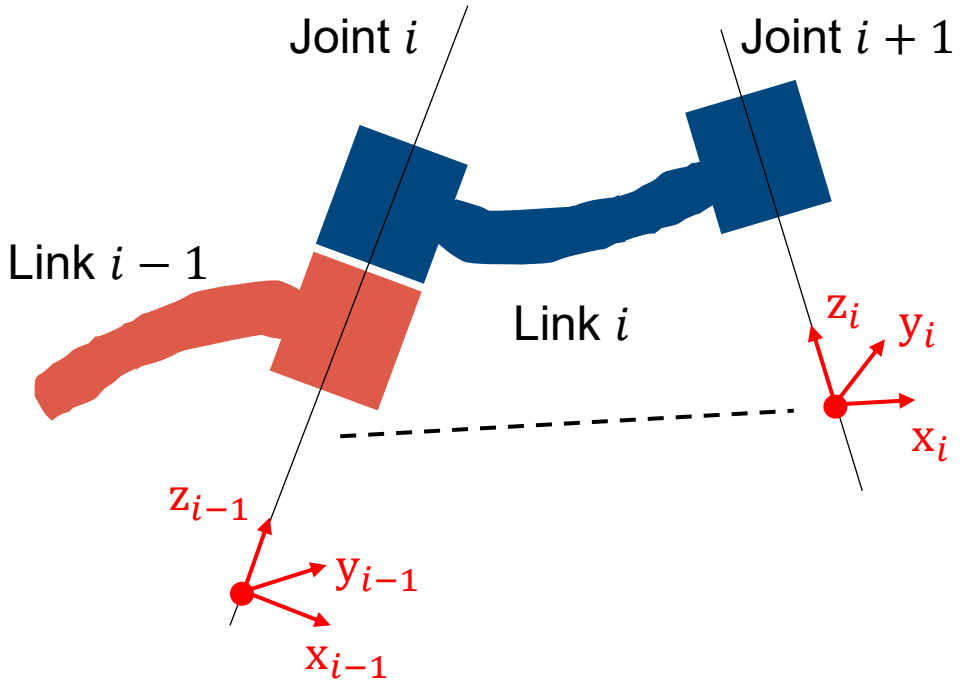
The standard notation presented by Denavit and Hartenberg

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Comparison between modified and standard DH



Modified DH - proximal



Standard DH - distal

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Standard DH – Distal notation

- The coordinate frame of link $i-1$ is located on axis i instead of on axis $i-1$
- The frame in placed after the link – “distal”
- The z_{i-1} axis is along the motion of joint i

- a_i : the distance from z_{i-1} to z_i measured along x_i
- α_i : the angle between z_{i-1} and z_i measured about x_i
- d_i : the distance from x_{i-1} to x_i measured along z_{i-1}
- θ_i : the angle between x_{i-1} to x_i measured about z_{i-1}

- Refer to the Robot Academy video for standard DH notation (https://robotacademy.net.au/lesson/denavit-hartenberg-notation/)

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Comparison between modified and standard DH

Proximal or Modified DH

	z_{i-1} to z_i | x_{i-1}		x_{i-1} to x_i | z_i	
i	$\mathbf{a}_{i-1}$	$\boldsymbol{\alpha}_{i-1}$	$\mathbf{d}_i$	$\boldsymbol{\theta}_i$
1				
2				

$$^{i-1}T_i = \mathrm{R}_{x_{i-1}}(\alpha_{i-1})\mathrm{t}_{x_{i-1}}(\mathrm{a}_{i-1})\mathrm{R}_{z_i}(\theta_i)\mathrm{t}_{z_i}(\mathrm{d}_i)$$

$$= \begin{bmatrix} c\theta_i & -s\theta_i & 0 & \mathrm{a}_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}\mathrm{d}_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}\mathrm{d}_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Distal or Standard DH

	z_{i-1} to z_i | x_i		x_{i-1} to x_i | z_{i-1}	
i	$\mathbf{a}_i$	$\boldsymbol{\alpha}_i$	$\mathbf{d}_i$	$\boldsymbol{\theta}_i$
1				
2				

$$^{i-1}T_i = \mathrm{R}_{z_{i-1}}(\theta_i)\mathrm{t}_{z_{i-1}}(\mathrm{d}_i)\mathrm{t}_{x_i}(\mathrm{a}_i)\mathrm{R}_{x_i}(\alpha_i)$$

$$= \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & c\theta_i \mathrm{a}_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & s\theta_i \mathrm{a}_i \\ 0 & s\alpha_i & c\alpha_i & \mathrm{d}_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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There are other variations of the DH notation

Refer to the section on Geometric Representation from the Springer Handbook of Robotics for a comparison of the variations.

Reference:

Waldron, K., Schmiedeler, J. (2008). Kinematics. In: Siciliano, B., Khatib, O. (eds) Springer Handbook of Robotics. Springer, Berlin, Heidelberg.

https://doi.org/10.1007/978-3-540-30301-5_2

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Why are 4 parameters sufficient?

- In general, 6 parameters are required to relate the pose of two rigid bodies in 3D space (3 rotation and 3 translation).
- But the set of 4 DH parameters completely describes the relation between adjacent links.

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Why are 4 parameters sufficient?

- In general, 6 parameters are required to relate the pose of two rigid bodies in 3D space (3 rotation and 3 translation).
- But the set of 4 DH parameters completely describes the relation between adjacent links.
- This is because the DH notation imposes constraints on where the coordinate frames are placed.

In the modified DH notation,

x_i intersects z_{i+1}

x_i is perpendicular to z_{i+1}

In the standard DH notation,

x_i intersects z_{i-1}

x_i is perpendicular to z_{i-1}

Consequence: the link frame is not necessarily on the link

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We desire a kinematic model to have three features

1. Completeness: all the spatial geometries of successive joints in a kinematic chain must be describable.
2. Continuity: small changes in the spatial geometry of joints must result in small changes of the describing parameters.
3. Minimality: the model must include only a minimal number of parameters.

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URDF notation

- Unified Robot Description Format
- Uses an XML (extensible markup language) type notation to represent the kinematic structure of a robot
- Refer to this document from Peter Corke for an excellent introduction https://petercorke.com/matlab/understanding-urdf-with-matlab/

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A snippet of URDF for a planar robot

```
1 <?xml version="1.0"?>
2 <robot name="planar2">
3
4               <link name="base_link" />
5               <link name="link1" />
6               <link name="link2" />
7               <link name="end" />
8
9               <joint name="q1" type="continuous">
10                              <parent link="base_link" />
11                              <child link="link1" />
12                              <axis xyz="0 0 1" />
13              </joint>
14
15              <joint name="q2" type="continuous">
16                              <parent link="link1" />
17                              <child link="link2" />
18                              <origin xyz="1 0 0" />
19                              <axis xyz="0 0 1" />
20              </joint>
21
22              <joint name="notajoint" type="fixed">
23                              <parent link="link2" />
24                              <child link="end" />
25                              <origin xyz="1 0 0" />
26              </joint>
27
28 </robot>
```

From https://petercorke.com/matlab/understanding-urdf-with-matlab/

URDF can also describe visual appearance

```
1 <?xml version="1.0"?>
2 <robot name="myfirst">
3
4       <link name="base_link">
5             <visual>
6                   <geometry>
7                         <cylinder length="0.6" radius="0.2"/>
8                   </geometry>
9             </visual>
10      </link>
11 </robot>
```

From https://petercorke.com/matlab/understanding-urdf-with-matlab/

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Summary

- There are several methods to assign frames to the links of a serial manipulator.
- The goal is to be able to describe the pose of the end effector with respect to a known base frame.
- No matter the method used, the transformation matrix must completely define the configuration of the manipulator arm.

- Today's practical – forward kinematics theory and practice with CoppeliaSim.
- What can you do if you know the forward kinematics of a manipulator?

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Next week

- We will discuss the opposite problem – to what position should each joint be moved so that the end effector can reach a desired pose?
- Inverse kinematics

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That’s all for today!

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